

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

Key relations :

- For a device connected directly across the battery , the voltage across it equals the battery voltage : $V = 12\text{ V}$.
- The power dissipated by the bulb is $P = V \cdot I$, so the current is $I = P/V = 40\text{ W} / 12\text{ V} = 3.33\text{ A}$.
- Using Ohm's law , the resistance is $R = V/I = 12\text{ V} / 3.33\text{ A} \approx 3.6\text{ }\Omega$.

Thus , the current is 3.33 amps and the required resistance is 3.6 ohms . Correct answer : E . 3.33 amps , 3.6 ohms .

</think>

<answer>

E . 3.33 amps , 3.6 ohms

</answer>